

A Technical Introduction to Semantic segmentation

Section 1

The simplest method of image segmentation is called the thresholding method. This method is based on a clip-level (or a threshold value) to turn a gray-scale image into a binary image. The key of this method is to select the threshold value (or values when multiple-levels are selected). Several popular methods are used in industry including the maximum entropy method, balanced histogram thresholding, Otsu's method (maximum variance), and k-means clustering. Recently, methods have been developed for thresholding computed tomography (CT) images. The key idea is that, unlike Otsu's method, the thresholds are derived from the radiographs instead of the (reconstructed) image. New methods suggest the use of multi-dimensional, fuzzy rule-based, non-linear thresholds. In these approaches, the decision regarding each pixel's membership in a segment is based on multi-dimensional rules derived from fuzzy logic and evolutionary algorithms, considering factors such as image lighting, environment, and application.

Section 2

Fast marching methods The fast marching method has been used in image segmentation, and this model has been improved (permitting both positive and negative propagation speeds) in an approach called the generalized fast marching method.

Section 3

Simulated annealing (SA) Derived as an analogue of annealing in metallurgy, simulated annealing (SA) uses change in pixel label over iterations and estimates the difference in energy of each newly formed graph to the initial data. If the newly formed graph is more profitable, in terms of low energy cost, given by:

Section 4

$$\operatorname{argmin}_{u,K} \gamma |K| + \mu \int K C |\nabla u|^2 dx + \int (u-f)^2 dx.$$

Section 5

Compression-based methods Compression based methods postulate that the optimal segmentation is the one that minimizes, over all possible segmentations, the coding length of the data. The connection between these two concepts is that segmentation tries to find patterns in an image and any regularity in the image can be used to compress it. The method describes each segment by its texture and boundary shape. Each of these components is modeled by a probability distribution function and its coding

length is computed as follows:

Section 6

Level-set methods The level-set method was initially proposed to track moving interfaces by Dervieux and Thomasset in 1979 and 1981 and was later reinvented by Osher and Sethian in 1988. This has spread across various imaging domains in the late 1990s. It can be used to efficiently address the problem of curve/surface/etc. propagation in an implicit manner. The central idea is to represent the evolving contour using a signed function whose zero corresponds to the actual contour. Then, according to the motion equation of the contour, one can easily derive a similar flow for the implicit surface that when applied to the zero level will reflect the propagation of the contour. The level-set method affords numerous advantages: it is implicit, is parameter-free, provides a direct way to estimate the geometric properties of the evolving structure, allows for change of topology, and is intrinsic. It can be used to define an optimization framework, as proposed by Zhao, Merriman and Osher in 1996. One can conclude that it is a very convenient framework for addressing numerous applications of computer vision and medical image analysis. Research into various level-set data structures has led to very efficient implementations of this method.

Section 7

This mathematical expression can be implemented by convolving with an appropriate mask. If we extend this equation to three dimensions (x,y,z), the intensity at each pixel location around a central pixel at (x, y, z) is replaced by their corresponding values. This equation becomes particularly useful when we assume that all pixels have unit spacing along each axis. A sphere mask has been developed for use with three-dimensional datasets. The sphere mask is designed to use only integer arithmetic during calculations, thereby eliminating the need for floating-point hardware or software. When applying these concepts to actual images represented as arrays of numbers, we need to consider what happens when we reach an edge or border region. The function $g(x, y)$ is defined as:

Section 8

Isolated Point Detection The detection of isolated points in an image is a fundamental part of image segmentation. This process primarily depends on the second derivative, indicating the use of the Laplacian operator. The Laplacian of a function $f(x, y)$ is given by:

Section 9

$$\alpha (1 - \delta(\ell_i - \ell_{\text{initial}})) + \beta \sum_{q \in N(i)} (1 - \delta(\ell_i - \ell_q))$$

$$\sum_i \left(\sum_{j \in N(i)} \alpha |q(i) - q(j)| + \beta \sum_{j \in N(i)} \delta_{ij} |q(i) - q(j)| \right)$$

Section 10

Multi-scale segmentation Image segmentations are computed at multiple scales in scale space and sometimes propagated from coarse to fine scales; see scale-space segmentation. Segmentation criteria can be arbitrarily complex and may take into account global as well as local criteria. A common requirement is that each region must be connected in some sense.

Section 11

where α is the penalty for change in pixel label and β is the penalty for difference in label between neighboring pixels and chosen pixel. Here $N(i)$ is neighborhood of pixel i and δ is the Kronecker delta function. A major issue with ICM is that, similar to gradient descent, it has a tendency to rest over local maxima and thus not obtain a globally optimal labeling scheme.

Section 12

Partial differential equation-based methods Using a partial differential equation (PDE)-based method and solving the PDE equation by a numerical scheme, one can segment the image. Curve propagation is a popular technique in this category, with numerous applications to object extraction, object tracking, stereo reconstruction, etc. The central idea is to evolve an initial curve towards the lowest potential of a cost function, where its definition reflects the task to be addressed. As for most inverse problems, the minimization of the cost functional is non-trivial and imposes certain smoothness constraints on the solution, which in the present case can be expressed as geometrical constraints on the evolving curve.